CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 14: Equation of a Line

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Forms of the equation

1. $ABCD$ is a parallelogram with vertices $A(3,3)$, $B(0,-2)$, $C(-4,-2)$ and $D(-1,3)$, read from a graph. (a) Find the point P where the diagonals meet. (b) Find the slopes of CB and DA and check that these sides are parallel. (c) Find the equation of the diagonal AC. ICSE 2025, 5 marks

SOLUTION

(a) The diagonals of a parallelogram bisect each other, so P is the mid-point of AC: $P = \left(\frac{3-4}{2}, \frac{3-2}{2}\right) = \left(-\frac{1}{2}, \frac{1}{2}\right)$.

(b) Slope of $CB = \frac{-2-(-2)}{0-(-4)} = 0$ and slope of $DA = \frac{3-3}{3-(-1)} = 0$. The slopes are equal, so $CB \parallel DA$.

(c) Slope of $AC = \frac{3-(-2)}{3-(-4)} = \frac{5}{7}$. Using $A(3,3)$: $y-3 = \frac{5}{7}(x-3)$, so $7y-21 = 5x-15$.

Answer: (a) $P\left(-\frac{1}{2}, \frac{1}{2}\right)$ **(b) both slopes are 0 (c)** $5x-7y+6=0$

2. Triangle ABC has vertices $A(1,-1)$, $B(0,4)$ and $C(-6,4)$, and D is the mid-point of BC. Find (a) the coordinates of D (b) the equation of the median AD. ICSE 2023, 3 marks

SOLUTION

(a) $D = \left(\frac{0-6}{2}, \frac{4+4}{2}\right) = (-3,4)$

(b) Slope of $AD = \frac{4-(-1)}{-3-1} = -\frac{5}{4}$. Through A: $y+1 = -\frac{5}{4}(x-1)$, so $4y+4 = -5x+5$.

Answer: (a) $D(-3,4)$ **(b)** $5x+4y-1=0$

3. $A(2,5)$, $B(-1,2)$ and $C(5,8)$ are the vertices of a triangle. M is the point on AB with $AM : MB = 1:2$. Find M, and then the equation of the line through C and M. ICSE 2018, 4 marks

SOLUTION

$M = \left(\frac{1(-1)+2(2)}{3}, \frac{1(2)+2(5)}{3}\right) = (1,4)$

Slope of $CM = \frac{8-4}{5-1} = 1$, so $y-4 = 1(x-1)$.

Answer: $M(1,4)$**; the line is** $y = x+3$

2 Lines parallel or perpendicular to a given line

4. Which line passes through the origin and is parallel to $3x + 4y + 7 = 0$? (a) $3x + 4y + 5 = 0$ (b) $4x - 3y - 5 = 0$ (c) $4x - 3y = 0$ (d) $3x + 4y = 0$ ICSE 2025, 1 mark

SOLUTION

A parallel line has the same x and y coefficients: $3x + 4y + k = 0$.

It passes through $(0, 0)$, so $k = 0$.

Answer: (d)

5. The line PQ meets the positive x-axis at P, 5 units from the origin, and the negative y-axis at Q, 3 units from the origin. The line RS passes through the origin and is perpendicular to PQ. Find (a) the coordinates of P and Q (b) the equation of RS. ICSE 2025, 3 marks

SOLUTION

(a) $P(5, 0)$ and $Q(0, -3)$.

(b) Slope of $PQ = \dfrac{0 - (-3)}{5 - 0} = \dfrac{3}{5}$, so the slope of RS is $-\dfrac{5}{3}$.

RS passes through the origin: $y = -\dfrac{5}{3}x$.

Answer: (a) $P(5, 0)$, $Q(0, -3)$ **(b)** $5x + 3y = 0$

6. (a) The lines $kx - y + 4 = 0$ and $2y = 6x + 7$ are perpendicular. Find k. (b) Find the equation of the line parallel to $2y = 6x + 7$ that passes through $(-1, 1)$. ICSE 2024, 3 marks

SOLUTION

(a) The first line is $y = kx + 4$, with slope k. The second is $y = 3x + \dfrac{7}{2}$, with slope 3.

Perpendicular lines: $k \times 3 = -1$, so $k = -\dfrac{1}{3}$.

(b) A parallel line has slope 3: $y - 1 = 3(x + 1)$

Answer: (a) $k = -\dfrac{1}{3}$ **(b)** $y = 3x + 4$

7. The lines $5x - 3y + 2 = 0$ and $6x - py + 7 = 0$ are perpendicular. Find p. Then find the equation of the line through $(-2, -1)$ parallel to $6x - py + 7 = 0$. ICSE 2020, 3 marks

SOLUTION

Slopes: $\dfrac{5}{3}$ and $\dfrac{6}{p}$. Perpendicular: $\dfrac{5}{3} \times \dfrac{6}{p} = -1$, so $p = -10$.

The second line is $6x + 10y + 7 = 0$, with slope $-\dfrac{3}{5}$.

Parallel line through $(-2, -1)$: $y + 1 = -\dfrac{3}{5}(x + 2)$, so $5y + 5 = -3x - 6$.

Answer: $p = -10$**; the line is** $3x + 5y + 11 = 0$

8. The vertices of $\triangle ABC$ are $A(3, 8)$, $B(-1, 2)$ and $C(6, -6)$. Find (a) the slope of BC (b) the equation of the line through A perpendicular to BC. ICSE 2019, 3 marks

SOLUTION

(a) Slope of $BC = \dfrac{-6-2}{6-(-1)} = -\dfrac{8}{7}$

(b) The perpendicular line has slope $\dfrac{7}{8}$: $y - 8 = \dfrac{7}{8}(x-3)$, so $8y - 64 = 7x - 21$.

Answer: (a) $-\dfrac{8}{7}$ **(b)** $7x - 8y + 43 = 0$

9. The lines $3x - 5y = 7$ and $4x + ay + 9 = 0$ are perpendicular. Find a. ICSE 2018, 3 marks

SOLUTION

Slopes: $\dfrac{3}{5}$ and $-\dfrac{4}{a}$. Perpendicular: $\dfrac{3}{5} \times \left(-\dfrac{4}{a}\right) = -1$

$12 = 5a$

Answer: $a = \dfrac{12}{5}$

10. $A(-1, 3)$, $B(4, 2)$ and $C(3, -2)$ are the vertices of a triangle. Find (a) the centroid G (b) the equation of the line through G parallel to AC. ICSE 2017, 3 marks

SOLUTION

(a) $G = \left(\dfrac{-1+4+3}{3}, \dfrac{3+2-2}{3}\right) = (2, 1)$

(b) Slope of $AC = \dfrac{-2-3}{3-(-1)} = -\dfrac{5}{4}$. The line is $y - 1 = -\dfrac{5}{4}(x-2)$, so $4y - 4 = -5x + 10$.

Answer: (a) $G(2, 1)$ **(b)** $5x + 4y - 14 = 0$

3 Inclination and slope

11. The points $A(x, y)$, $B(3, -2)$ and $C(4, -5)$ are collinear. Write y in terms of x. (a) $3x - 11$ (b) $11 - 3x$ (c) $3x - 7$ (d) $7 - 3x$ ICSE 2024, 1 mark

SOLUTION

Slope of $BC = \dfrac{-5+2}{4-3} = -3$. Collinear points give the same slope for AB.

$\dfrac{y+2}{x-3} = -3$, so $y = -3x + 9 - 2 = 7 - 3x$

Answer: (d)

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